



**RV College of
Engineering®**
Mysore Road, RV Vidyaniketan f
Bengaluru - 560059, Karnataka,

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Machine Learning (MLOps) Tutorial

Real-Time UAV Detection and Tracking on Mobile Platform Using CNN

Team No 5

Name	USN
Sachit Ramesha Gowda	1RV23AI084
Ravi Kishan Kumar	1RV23AI080
Rayala Yuvaraj Vaishnav	1RV23AI081
S Vishwanatha	1RV23AI083



Introduction



- UAVs (drones) are increasingly used in civilian and defense domains
- Detecting and tracking UAVs in real-time is critical for surveillance and security
- AI-based computer vision enables automatic UAV detection from images and videos
- This project integrates CNN-based detection with a complete MLOps pipeline



Motivation

- Rising drone threats in restricted and sensitive areas
- Need for real-time, accurate, and deployable UAV detection systems
- Traditional ML models fail without proper deployment, monitoring, and versioning
- MLOps ensures:
 - Reproducibility
 - Scalability
 - Continuous monitoring





Problem Definition

Existing UAV datasets are largely single-class, which limits model generalization across different environments and conditions.

This creates challenges in maintaining accurate and reliable UAV detection in real-world scenarios.

Therefore, accurate bounding box detection, real-time inference on mobile platforms, and automated ML lifecycle management using MLOps are required.



Objectives

- Build a real-time UAV detection model using YOLO architecture
- Perform bounding box detection and object tracking
- Apply data augmentation for better generalization
- Implement a complete MLOps pipeline:
 - Versioning
 - Experiment tracking
 - Deployment
 - Monitoring



Data set Description

- Dataset size: 51.4k UAV images (~5 GB)
- Single-class dataset (UAV only)
- Used for UAV detection and tracking research

Features included:

- Width
- Height
- UAV class label
- Object ID
- Bounding box area



Risk Analysis and Mitigation

Risk Analysis:

- False positives (birds detected as UAVs)
- Overfitting due to single-class data
- A model can be poisoned if it repeatedly learns from UAV images captured in similar backgrounds, colours, or environments, causing it to associate background patterns instead of actual UAV features. This can lead to biased predictions, false positives, or intentional misclassification when the background is manipulated.

Risk Mitigation:

- Data augmentation
- Drift monitoring using MLOps tools
- Continuous retraining



Phase II – MLOPs Life Cycle

	Component	MLOPs Tools
1	Data Management	Roboflow
2	Version Control	Hugging Face and Git
3	Experiment Tracking & Model Registry	Weights and Biases (W&B)
4	Automation and Pipeline Orchestration	ZenML
5	Model Deployment & Serving	Nix OS and ONNX Runtime
6	Monitoring & Drift Detection	Evidently AI

Roboflow - Data Labeling



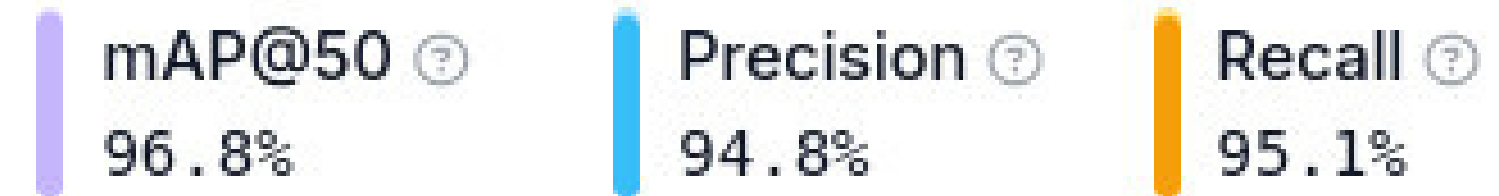
About Roboflow

- Roboflow is a computer vision–specific data management platform that simplifies the entire data preparation process.
- You can integrate Roboflow at various stages of your development pipeline using their APIs and SDKs, or utilize its end-to-end interface to manage the process from image collection to inference.

Why Roboflow?

- Roboflow was used to efficiently label and manage UAV image datasets with consistent bounding box annotations.

METRICS



Outputs per training example: 3

- Flip: Horizontal, Vertical
- Crop: 0% Minimum Zoom, 20% Maximum Zoom
- Shear: $\pm 10^\circ$ Horizontal, $\pm 13^\circ$ Vertical
- Saturation: Between -25% and +25%
- Exposure: Between -9% and +9%
- Blur: Up to 2.5px



Hugging Face - Data Versioning

About Hugging Face

Hugging Face Hub supports:

- Dataset hosting & versioning
- Model versioning
- Artifact storage
- Access control
- Easy sharing & reuse



Hugging Face

Why Hugging Face?

- Centralized ML Artifact Management
 - Models + datasets in one place
 - Easy collaboration
- Git-Based Versioning (Under the Hood)
 - Each dataset/model is a Git repo
 - Commit history
 - Branching and tags

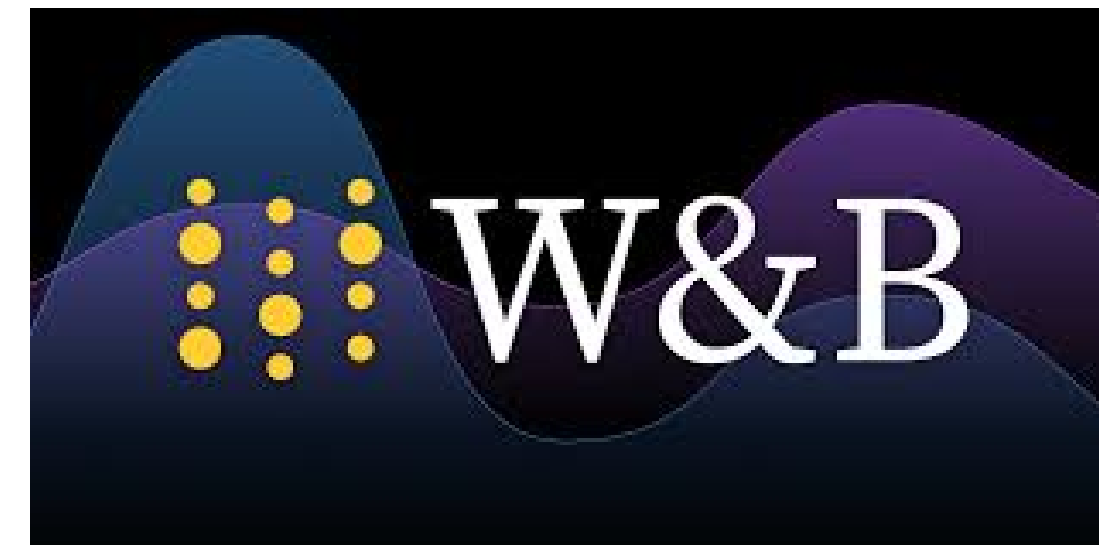


git

W&B - Experiment Tracking & Model Registry

About W&B

- Weights & Biases is a cloud-based MLOps tool used to track and visualize machine learning training processes.
- It logs model metrics and training configurations in real time and displays them through an interactive web dashboard.

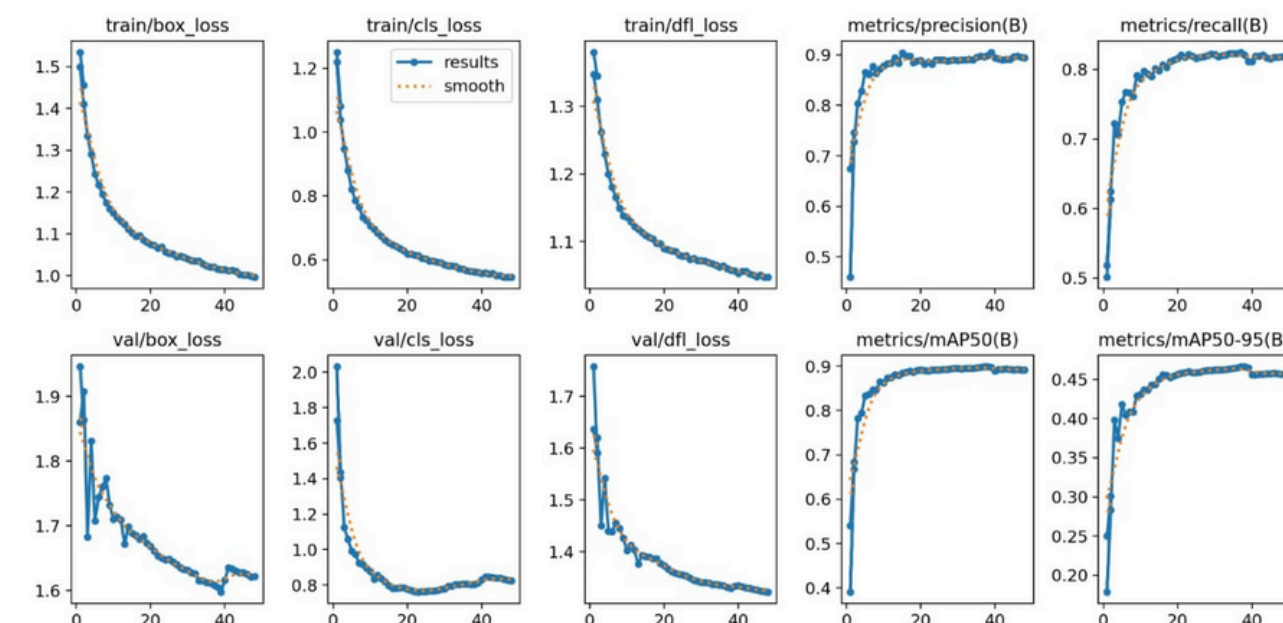


Hyperparameters Used:

- learning rate
- weight decay
- optimizer (adamw)
- momentum
- weight_decay
- warmup_epochs
- warmup_momentum
- warmup_bias_lr

Why W&B?

- To track and log training metrics such as loss, accuracy, and detection confidence
- To record hyperparameters for each training run
- To manage model versions using W&B Model Registry for reproducibility





ZenML - Pipeline Orchestration

About ZenML

- ZenML is an open-source ML Ops framework that provides a layer of abstraction for building and orchestrating machine learning pipelines.
- Rather than being a standalone orchestrator, ZenML functions as a "meta-orchestrator, allowing you to define your pipelines in pure Python and then run them on a variety of existing orchestration backends.

Why ZenML?

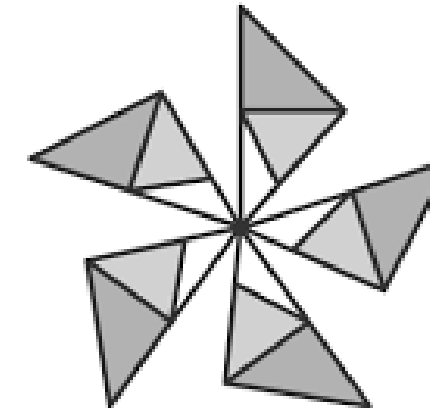
ZenML is ML-native, unlike general workflow tools.

Advantages:

- Step-based pipeline abstraction
- Tool-agnostic integration
- Designed specifically for MLOps
- Lightweight and easy to explain



NixOS + ONNX - Model Deployment



ONNX
RUNTIME

About Nix OS

- Weights & Biases (WandB) is a machine learning development platform that allows users to track and visualize various aspects of their model training process in real-time.
- Track model performance metrics such as accuracy, loss, and other evaluation metrics during the training and evaluation phases.
- Visualize the model's learning process using graphs, charts, and histograms to gain insights into how the model is performing.

Why Nix OS?

NixOS enables:

- Deterministic reproducibility
- System drift is avoided
- Minimal dependency issues



NixOS



Evidently AI - Drift Detection

About Evidently AI

- Evidently AI is an open-source monitoring and evaluation tool used in MLOps to analyze machine learning models in production.
- It helps track data quality, model performance, and detect data or prediction drift over time using automated reports and dashboards.



Why Evidently AI?

- To monitor the UAV detection model after deployment
- To detect data drift caused by changes in environment, lighting, background, or camera conditions
- To identify performance degradation of the model over time
- To ensure long-term reliability and robustness of real-time UAV detection

Tools and Techniques Used

Techniques Used

Convolutional Neural Networks (CNNs)

Used as the foundational architecture for extracting spatial features from UAV images.

YOLOv12 Object Detection

Enables real-time UAV detection with bounding box localization and high inference speed.

Bounding Box Detection

Identifies and localizes UAVs within images and video frames.

Data Augmentation

Techniques such as rotation, scaling, flipping, brightness adjustment, and noise addition are used to improve generalization.

Tools and Techniques Used

Techniques Used

Sliding Window Technique

Breaks high-resolution video frames into smaller regions to improve detection accuracy.

ID-based Object Tracking

Tracks UAVs across consecutive video frames for continuous monitoring.

Model Export using ONNX

Converts trained models into an optimized format for faster inference during deployment.